Device Comfort and Usability Survey: Patient/Client

This survey aims to evaluate the comfort, usability, and overall user experience of the stroke rehabilitation device. There are no right or wrong answers. Your honest feedback will help improve the design and usability of the device. All responses will be kept confidential.

Participant Information

Name: Ling Wei Hsuen
Age: 23
Gender: Female
Occupation: Student
Date of Test: 2nd April 2026
Tester Name: Nadia and Ron
Total Time Using Device: 40 mins

Section A: General Usability

Please rate the following statements on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

I felt confident using the device. Rating (1–5): 5 Comments: _____
I would be willing to use this device regularly Rating (1–5): 5 Comments: _____

Section B: Setup, Adjustment and Handling

The use process was easy to understand. Rating (1–5): 5 Comments: _____
The device was easy to adjust to fit my leg. Rating (1–5): 5 Comments: _____
It was easy to get into and out of the device. Rating (1–5): 5 Comments: _____

Section C: Fit and Physical Comfort

The padding and straps were comfortable. Rating (1–5): 5 Comments: _____
The device fit my leg well. Rating (1–5): 5 Comments: _____
I did not experience pressure points or pain. Rating (1–5): 5 Comments: _____
The device supported my leg securely. Rating (1–5): 5 Comments: _____
The device accommodated my posture comfortably. Rating (1–5): 5 Comments: _____

Section D: Exercise Experience and Effectiveness

Passive exercises felt smooth and comfortable. Rating (1–5): 5 Comments: _____
Active exercises felt natural and controlled. Rating (1–5): 5 Comments: _____
The range of motion felt appropriate. Rating (1–5): 5 Comments: _____
I did not feel excessive strain during exercises. Rating (1–5): 5 Comments: _____
The device helped me perform exercises effectively. Rating (1–5): 5 Comments: _____

Section E: Safety and Confidence

I felt safe while using the device. Rating (1–5): 5 Comments: _____
The device felt stable during movement. Rating (1–5): 5 Comments: _____
The emergency stop and help button was easy to reach and use. Rating (1–5): 5 Comments: _____

Section F: Device Design and Practicality

The size of the device was appropriate. Rating (1–5): 5 Comments: _____
The device did not feel too bulky. Rating (1–5): 5 Comments: _____

Section G: Effort and Workload

Using the device required too much physical effort. Rating (1–5): 1 Comments: _____
I felt comfortable completing exercises. Rating (1–5): 5 Comments: _____

Section G: Overall Feedback

What did you like most about the device? It was easy to get in and it allows me to warm up. Comfortable and feels safe to use, especially with the emergency stop machine.
What did you find uncomfortable or difficult? It was slightly uncomfortable that there wasn't a back rest, however, when this is actually used for patients, there would be one.
What improvements would you suggest? NIL

Would you be willing to use this device regularly if recommended by a therapist? (Yes / No)

Remarks: This survey is an adapted hybrid usability survey based on recognised standards and validated constructs, refined using current lower-limb rehabilitation-device literature.

References

[1] J. Brooke, “SUS: A quick and dirty usability scale,” in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and A. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[2] L. Demers, R. Weiss-Lambrou, and B. Ska, “The Quebec User Evaluation of Satisfaction with Assistive Technology (QUEST 2.0): An overview and recent progress,” *Disability and Rehabilitation*, vol. 24, no. 1–3, pp. 21–30, 2002.

[3] S. G. Hart and L. E. Staveland, “Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research,” in *Advances in Psychology*, vol. 52, P. A. Hancock and N. Meshkati, Eds. Amsterdam, Netherlands: North-Holland, 1988, pp. 139–183.

[4] ISO 62366-1, *Medical devices—Application of usability engineering to medical devices—Part 1: Application of usability engineering to medical devices*, Geneva, Switzerland: International Organization for Standardization, 2015.

[5] ISO 62366-1 Amendment 1, *Medical devices—Application of usability engineering to medical devices—Amendment 1*, Geneva, Switzerland: International Organization for Standardization, 2020.

[6] D. R. Louie and J. J. Eng, “Powered robotic exoskeletons in post-stroke rehabilitation of gait: A scoping review,” *Journal of NeuroEngineering and Rehabilitation*, vol. 13, no. 53, 2016.

Device Comfort and Usability Survey: Patient/Client

This survey aims to evaluate the comfort, usability, and overall user experience of the stroke rehabilitation device. There are no right or wrong answers. Your honest feedback will help improve the design and usability of the device. All responses will be kept confidential.

Participant Information

Name: Chloe Wong
Age: 22
Gender: Female
Occupation: Student
Date of Test: 3 Apr 2026
Tester Name: Nadia and Ron
Total Time Using Device: 15 minutes

Section A: General Usability

Please rate the following statements on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

I felt confident using the device. Rating (1–5): 3 Comments: _____
I would be willing to use this device regularly Rating (1–5): 4 Comments: _____

Section B: Setup, Adjustment and Handling

The use process was easy to understand. Rating (1–5): 5 Comments: _____
It was easy to get into and out of the device. Rating (1–5): 4 Comments: _____

Section C: Fit and Physical Comfort

The padding and straps were comfortable. Rating (1–5): 4 Comments: _____
The device fit my leg well. Rating (1–5): 3 Comments: When stretching, it felt a bit long for my leg
I did not experience pressure points or pain. Rating (1–5): 4 Comments: _____
The device supported my leg securely. Rating (1–5): 3 Comments: Felt like my leg might come out when stretching out
The device accommodated my posture comfortably. Rating (1–5): 3 Comments: _____

Section D: Exercise Experience and Effectiveness

Passive exercises felt smooth and comfortable. Rating (1–5): 4 Comments: _____
Active exercises felt natural and controlled. Rating (1–5): 3 Comments: _____
The range of motion felt appropriate. Rating (1–5): 3 Comments: _____
I did not feel excessive strain during exercises. Rating (1–5): 4 Comments: _____
The device helped me perform exercises effectively. Rating (1–5): 4 Comments: _____

Section E: Safety and Confidence

I felt safe while using the device. Rating (1–5): 5 Comments: _____
The device felt stable during movement. Rating (1–5): 3 Comments: Device felt a bit shaky at times
The emergency stop and help button was easy to reach and use. Rating (1–5): 5 Comments: _____

Section F: Device Design and Practicality

The size of the device was appropriate. Rating (1–5): 4 Comments: _____
The device did not feel too bulky. Rating (1–5): 4 Comments: _____

Section G: Effort and Workload

Using the device required too much physical effort. Rating (1–5): 2 Comments: _____
I felt comfortable completing exercises. Rating (1–5): 4 Comments: _____

Section G: Overall Feedback

What did you like most about the device? UI was easy to understand and use, device was also easy to get into and use
What did you find uncomfortable or difficult? For stretching the leg, sometimes, it felt a bit too long for me, and I got a bit worried my foot would come out
What improvements would you suggest? Setup for the device could be more inclusive for people of different sizes and proportions

Would you be willing to use this device regularly if recommended by a therapist? (Yes / No) Yes

Remarks: This survey is an adapted hybrid usability survey based on recognised standards and validated constructs, refined using current lower-limb rehabilitation-device literature.

References

[1] J. Brooke, “SUS: A quick and dirty usability scale,” in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and A. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[2] L. Demers, R. Weiss-Lambrou, and B. Ska, “The Quebec User Evaluation of Satisfaction with Assistive Technology (QUEST 2.0): An overview and recent progress,” *Disability and Rehabilitation*, vol. 24, no. 1–3, pp. 21–30, 2002.

[3] S. G. Hart and L. E. Staveland, “Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research,” in *Advances in Psychology*, vol. 52, P. A. Hancock and N. Meshkati, Eds. Amsterdam, Netherlands: North-Holland, 1988, pp. 139–183.

[4] ISO 62366-1, *Medical devices—Application of usability engineering to medical devices—Part 1: Application of usability engineering to medical devices*, Geneva, Switzerland: International Organization for Standardization, 2015.

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[6] D. R. Louie and J. J. Eng, “Powered robotic exoskeletons in post-stroke rehabilitation of gait: A scoping review,” *Journal of NeuroEngineering and Rehabilitation*, vol. 13, no. 53, 2016.

Device Comfort and Usability Survey: Patient/Client

This survey aims to evaluate the comfort, usability, and overall user experience of the stroke rehabilitation device. There are no right or wrong answers. Your honest feedback will help improve the design and usability of the device. All responses will be kept confidential.

Participant Information

Name: Abigail Tan Huey Ern
Age: 22
Gender: Female
Occupation: Student
Date of Test: 3 Apr 2026
Tester Name: Nadia and Ron
Total Time Using Device:

Section A: General Usability

Please rate the following statements on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

I felt confident using the device. Rating (1–5): Comments: _____3_____
I would be willing to use this device regularly Rating (1–5): Comments: ____2_____

Section B: Setup, Adjustment and Handling

The use process was easy to understand. Rating (1–5): Comments: _____5_____
It was easy to get into and out of the device. Rating (1–5): Comments: _____4_____

Section C: Fit and Physical Comfort

The padding and straps were comfortable. Rating (1–5): Comments: _____4_____
The device fit my leg well. Rating (1–5): Comments: _____2_____
I did not experience pressure points or pain. Rating (1–5): Comments: _____5_____
The device supported my leg securely. Rating (1–5): Comments: _____4_____
The device accommodated my posture comfortably. Rating (1–5): Comments: _____2_____

Section D: Exercise Experience and Effectiveness

Passive exercises felt smooth and comfortable. Rating (1–5): Comments: _____3_____
Active exercises felt natural and controlled. Rating (1–5): Comments: _____4_____
The range of motion felt appropriate. Rating (1–5): Comments: _____4_____
I did not feel excessive strain during exercises. Rating (1–5): Comments: _____3_____
The device helped me perform exercises effectively. Rating (1–5): Comments: _____3_____

Section E: Safety and Confidence

I felt safe while using the device. Rating (1–5): Comments: _____3_____
The device felt stable during movement. Rating (1–5): Comments: _____1_____
The emergency stop and help button was easy to reach and use. Rating (1–5): Comments: _____5_____

Section F: Device Design and Practicality

The size of the device was appropriate. Rating (1–5): Comments: _____3_____
The device did not feel too bulky. Rating (1–5): Comments: _____5_____

Section G: Effort and Workload

Using the device required too much physical effort. Rating (1–5): Comments: _____2_____
I felt comfortable completing exercises. Rating (1–5): Comments: _____4_____

Section G: Overall Feedback

What did you like most about the device? It allowed a good amount of extension.
What did you find uncomfortable or difficult? I felt like I was gonna break the device with the weight of my leg.
What improvements would you suggest? More sturdy materials and a chair for the patient to more comfortably put their leg at the right angle.

Would you be willing to use this device regularly if recommended by a therapist? (Yes / No) Yes

Remarks: This survey is an adapted hybrid usability survey based on recognised standards and validated constructs, refined using current lower-limb rehabilitation-device literature.

References

[1] J. Brooke, “SUS: A quick and dirty usability scale,” in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and A. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[2] L. Demers, R. Weiss-Lambrou, and B. Ska, “The Quebec User Evaluation of Satisfaction with Assistive Technology (QUEST 2.0): An overview and recent progress,” *Disability and Rehabilitation*, vol. 24, no. 1–3, pp. 21–30, 2002.

[3] S. G. Hart and L. E. Staveland, “Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research,” in *Advances in Psychology*, vol. 52, P. A. Hancock and N. Meshkati, Eds. Amsterdam, Netherlands: North-Holland, 1988, pp. 139–183.

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[5] ISO 62366-1 Amendment 1, *Medical devices—Application of usability engineering to medical devices—Amendment 1*, Geneva, Switzerland: International Organization for Standardization, 2020.

[6] D. R. Louie and J. J. Eng, “Powered robotic exoskeletons in post-stroke rehabilitation of gait: A scoping review,” *Journal of NeuroEngineering and Rehabilitation*, vol. 13, no. 53, 2016.

Device Comfort and Usability Survey: Patient/Client

This survey aims to evaluate the comfort, usability, and overall user experience of the stroke rehabilitation device. There are no right or wrong answers. Your honest feedback will help improve the design and usability of the device. All responses will be kept confidential.

Participant Information

Name: HUANG ZICHEN
Age: 19
Gender: Male
Occupation: Student
Date of Test: 3 Apr 2026
Tester Name: Nadia and Ron
Total Time Using Device: 15mins

Section A: General Usability

Please rate the following statements on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

I felt confident using the device. Rating (1–5): 5 Comments: _____
I would be willing to use this device regularly Rating (1–5): 5 Comments: _____

Section B: Setup, Adjustment and Handling

The use process was easy to understand. Rating (1–5): 5 Comments: _____
It was easy to get into and out of the device. Rating (1–5): 5 Comments: _____

Section C: Fit and Physical Comfort

The padding and straps were comfortable. Rating (1–5): 4 Comments: _____
The device fit my leg well. Rating (1–5): 5 Comments: _____
I did not experience pressure points or pain. Rating (1–5): 5 Comments: _____
The device supported my leg securely. Rating (1–5): 5 Comments: _____
The device accommodated my posture comfortably. Rating (1–5): 5 Comments: _____

Section D: Exercise Experience and Effectiveness

Passive exercises felt smooth and comfortable. Rating (1–5): 5 Comments: _____
Active exercises felt natural and controlled. Rating (1–5):5 Comments: _____
The range of motion felt appropriate. Rating (1–5):5 Comments: _____
I did not feel excessive strain during exercises. Rating (1–5): 5 Comments: _____
The device helped me perform exercises effectively. Rating (1–5): 5 Comments: _____

Section E: Safety and Confidence

I felt safe while using the device. Rating (1–5): 5 Comments: _____
The device felt stable during movement. Rating (1–5): 5 Comments: _____
The emergency stop and help button was easy to reach and use. Rating (1–5): 5 Comments: _____

Section F: Device Design and Practicality

The size of the device was appropriate. Rating (1–5): 5 Comments: _____
The device did not feel too bulky. Rating (1–5): 5 Comments: _____

Section G: Effort and Workload

Using the device required too much physical effort. Rating (1–5): 5 Comments: _____
I felt comfortable completing exercises. Rating (1–5): 5 Comments: _____

Section G: Overall Feedback

What did you like most about the device?
What did you find uncomfortable or difficult?
What improvements would you suggest?

Would you be willing to use this device regularly if recommended by a therapist? (Yes / No)

Remarks: This survey is an adapted hybrid usability survey based on recognised standards and validated constructs, refined using current lower-limb rehabilitation-device literature.

References

[1] J. Brooke, “SUS: A quick and dirty usability scale,” in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and A. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[2] L. Demers, R. Weiss-Lambrou, and B. Ska, “The Quebec User Evaluation of Satisfaction with Assistive Technology (QUEST 2.0): An overview and recent progress,” *Disability and Rehabilitation*, vol. 24, no. 1–3, pp. 21–30, 2002.

[3] S. G. Hart and L. E. Staveland, “Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research,” in *Advances in Psychology*, vol. 52, P. A. Hancock and N. Meshkati, Eds. Amsterdam, Netherlands: North-Holland, 1988, pp. 139–183.

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[6] D. R. Louie and J. J. Eng, “Powered robotic exoskeletons in post-stroke rehabilitation of gait: A scoping review,” *Journal of NeuroEngineering and Rehabilitation*, vol. 13, no. 53, 2016.

Device Comfort and Usability Survey: Patient/Client

This survey aims to evaluate the comfort, usability, and overall user experience of the stroke rehabilitation device. There are no right or wrong answers. Your honest feedback will help improve the design and usability of the device. All responses will be kept confidential.

Participant Information

Name: Shandrico Julian Lim
Age: 19
Gender: Male
Occupation: Student
Date of Test: 3 April 2026
Tester Name: Shandrico Julian Lim
Total Time Using Device: 15 minutes

Section A: General Usability

Please rate the following statements on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

I felt confident using the device. Rating (1–5): 5 Comments: _____
I would be willing to use this device regularly Rating (1–5): 5 Comments: _____

Section B: Setup, Adjustment and Handling

The use process was easy to understand. Rating (1–5): 5 Comments: _____
It was easy to get into and out of the device. Rating (1–5): 5 Comments: _____

Section C: Fit and Physical Comfort

The padding and straps were comfortable. Rating (1–5): 5 Comments: _____
The device fit my leg well. Rating (1–5): 5 Comments: _____
I did not experience pressure points or pain. Rating (1–5): 5 Comments: _____
The device supported my leg securely. Rating (1–5): 5 Comments: _____
The device accommodated my posture comfortably. Rating (1–5): 5 Comments: _____

Section D: Exercise Experience and Effectiveness

Passive exercises felt smooth and comfortable. Rating (1–5): 5 Comments: _____
Active exercises felt natural and controlled. Rating (1–5): 5 Comments: _____
The range of motion felt appropriate. Rating (1–5): 5 Comments: _____
I did not feel excessive strain during exercises. Rating (1–5): 5 Comments: _____
The device helped me perform exercises effectively. Rating (1–5): 5 Comments: _____

Section E: Safety and Confidence

I felt safe while using the device. Rating (1–5): 5 Comments: _____
The device felt stable during movement. Rating (1–5): 5 Comments: _____
The emergency stop and help button was easy to reach and use. Rating (1–5): 5 Comments: _____

Section F: Device Design and Practicality

The size of the device was appropriate. Rating (1–5): 5 Comments: _____
The device did not feel too bulky. Rating (1–5): 5 Comments: _____

Section G: Effort and Workload

Using the device required too much physical effort. Rating (1–5): 5 Comments: _____
I felt comfortable completing exercises. Rating (1–5): 5 Comments: _____

Section G: Overall Feedback

What did you like most about the device? It is easy to operate and intuitive
What did you find uncomfortable or difficult? It feels a bit shaky
What improvements would you suggest? Making it stable and if it can move left and right would be better

Would you be willing to use this device regularly if recommended by a therapist? (Yes / No) Yes

Remarks: This survey is an adapted hybrid usability survey based on recognised standards and validated constructs, refined using current lower-limb rehabilitation-device literature.

References

[1] J. Brooke, “SUS: A quick and dirty usability scale,” in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and A. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

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[6] D. R. Louie and J. J. Eng, “Powered robotic exoskeletons in post-stroke rehabilitation of gait: A scoping review,” *Journal of NeuroEngineering and Rehabilitation*, vol. 13, no. 53, 2016.

Device Comfort and Usability Survey: Patient/Client

This survey aims to evaluate the comfort, usability, and overall user experience of the stroke rehabilitation device. There are no right or wrong answers. Your honest feedback will help improve the design and usability of the device. All responses will be kept confidential.

Participant Information

Name: Reagan Moses Widjaja
Age: 20
Gender: Male
Occupation: Student
Date of Test: 3 Apr 2026
Tester Name: Nadia and Ron
Total Time Using Device: 15 minutes

Section A: General Usability

Please rate the following statements on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

I felt confident using the device. Rating (1–5): 5 Comments: __User Friendly_____
I would be willing to use this device regularly Rating (1–5): 5 Comments: _____

Section B: Setup, Adjustment and Handling

The use process was easy to understand. Rating (1–5): 5 Comments: _____
It was easy to get into and out of the device. Rating (1–5): 3 Comments: ___Need to lift leg up a lot to go in and out_____

Section C: Fit and Physical Comfort

The padding and straps were comfortable. Rating (1–5): 5 Comments: _____
The device fit my leg well. Rating (1–5): 5 Comments: _____
I did not experience pressure points or pain. Rating (1–5): 5 Comments: _____
The device supported my leg securely. Rating (1–5): 5 Comments: _____
The device accommodated my posture comfortably. Rating (1–5): 5 Comments: _____

Section D: Exercise Experience and Effectiveness

Passive exercises felt smooth and comfortable. Rating (1–5): 5 Comments: _____
Active exercises felt natural and controlled. Rating (1–5): 5 Comments: _____
The range of motion felt appropriate. Rating (1–5): 4 Comments: __Didn't fully stretch_____
I did not feel excessive strain during exercises. Rating (1–5): 5 Comments: _____
The device helped me perform exercises effectively. Rating (1–5): 5 Comments: _____

Section E: Safety and Confidence

I felt safe while using the device. Rating (1–5): 5 Comments: _____
The device felt stable during movement. Rating (1–5): 5 Comments: _____
The emergency stop and help button was easy to reach and use. Rating (1–5): 5 Comments: _____

Section F: Device Design and Practicality

The size of the device was appropriate. Rating (1–5): 5 Comments: _____
The device did not feel too bulky. Rating (1–5): 5 Comments: _____

Section G: Effort and Workload

Using the device required too much physical effort. Rating (1–5): 5 Comments: _____
I felt comfortable completing exercises. Rating (1–5): 5 Comments: _____

Section G: Overall Feedback

What did you like most about the device? Intuitive and gentle
What did you find uncomfortable or difficult? Putting the leg in and out was kinda awkward since it requires larger range of motion
What improvements would you suggest? Nothing much really

Would you be willing to use this device regularly if recommended by a therapist? (Yes / No) Yes

Remarks: This survey is an adapted hybrid usability survey based on recognised standards and validated constructs, refined using current lower-limb rehabilitation-device literature.

References

[1] J. Brooke, “SUS: A quick and dirty usability scale,” in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and A. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

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Device Comfort and Usability Survey: Patient/Client

This survey aims to evaluate the comfort, usability, and overall user experience of the stroke rehabilitation device. There are no right or wrong answers. Your honest feedback will help improve the design and usability of the device. All responses will be kept confidential.

Participant Information

Name: Sebastian Mc Eon Espina Besonia
Age: 25
Gender: M
Occupation: Student
Date of Test: 03/04/ 26
Tester Name: Nadia and Ron
Total Time Using Device: 15 minutes

Section A: General Usability

Please rate the following statements on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

I felt confident using the device. Rating (1–5): 4 Comments: _____
I would be willing to use this device regularly Rating (1–5): 5 Comments: _____

Section B: Setup, Adjustment and Handling

The use process was easy to understand. Rating (1–5): 5 Comments: _____
It was easy to get into and out of the device. Rating (1–5): 3 Comments: _____

Section C: Fit and Physical Comfort

The padding and straps were comfortable. Rating (1–5): 5 Comments: _____
The device fit my leg well. Rating (1–5): 5 Comments: _____
I did not experience pressure points or pain. Rating (1–5): 5 Comments: _____
The device supported my leg securely. Rating (1–5): 4 Comments: _____
The device accommodated my posture comfortably. Rating (1–5): 4 Comments: _____

Section D: Exercise Experience and Effectiveness

Passive exercises felt smooth and comfortable. Rating (1–5): 5 Comments: _____
Active exercises felt natural and controlled. Rating (1–5): 5 Comments: _____
The range of motion felt appropriate. Rating (1–5): 5 Comments: _____
I did not feel excessive strain during exercises. Rating (1–5): 4 Comments: _____
The device helped me perform exercises effectively. Rating (1–5): 5 Comments: _____

Section E: Safety and Confidence

I felt safe while using the device. Rating (1–5): 5 Comments: _____
The device felt stable during movement. Rating (1–5): 3 Comments: _____
The emergency stop and help button was easy to reach and use. Rating (1–5): 5 Comments: _____

Section F: Device Design and Practicality

The size of the device was appropriate. Rating (1–5): 5 Comments: _____
The device did not feel too bulky. Rating (1–5): 4 Comments: _____

Section G: Effort and Workload

Using the device required too much physical effort. Rating (1–5): 1 Comments: _____
I felt comfortable completing exercises. Rating (1–5): 5 Comments: _____

Section G: Overall Feedback

What did you like most about the device? I like how easy it was to operate
What did you find uncomfortable or difficult? I think the device needs to be a bit more secure
What improvements would you suggest? Add more mounting points for the structure

Would you be willing to use this device regularly if recommended by a therapist? (Yes / No) Yes

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References

[1] J. Brooke, “SUS: A quick and dirty usability scale,” in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and A. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[2] L. Demers, R. Weiss-Lambrou, and B. Ska, “The Quebec User Evaluation of Satisfaction with Assistive Technology (QUEST 2.0): An overview and recent progress,” *Disability and Rehabilitation*, vol. 24, no. 1–3, pp. 21–30, 2002.

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[6] D. R. Louie and J. J. Eng, “Powered robotic exoskeletons in post-stroke rehabilitation of gait: A scoping review,” *Journal of NeuroEngineering and Rehabilitation*, vol. 13, no. 53, 2016.

Device Comfort and Usability Survey: Patient/Client

This survey aims to evaluate the comfort, usability, and overall user experience of the stroke rehabilitation device. There are no right or wrong answers. Your honest feedback will help improve the design and usability of the device. All responses will be kept confidential.

Participant Information

Name: Ron
Age: 25
Gender: Male
Occupation: Student
Date of Test: 3 Apr 2026
Tester Name: Nadia
Total Time Using Device: 15 minutes

Section A: General Usability

Please rate the following statements on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

I felt confident using the device. Rating (1–5): 5 Comments: _____
I would be willing to use this device regularly Rating (1–5): 5 Comments: _____

Section B: Setup, Adjustment and Handling

The use process was easy to understand. Rating (1–5): 4 Comments: _____
It was easy to get into and out of the device. Rating (1–5): 3 Comments: _____

Section C: Fit and Physical Comfort

The padding and straps were comfortable. Rating (1–5): 4 Comments: _____
The device fit my leg well. Rating (1–5): 4 Comments: _____
I did not experience pressure points or pain. Rating (1–5): 5 Comments: _____
The device supported my leg securely. Rating (1–5): 4 Comments: _____
The device accommodated my posture comfortably. Rating (1–5): 4 Comments: _____

Section D: Exercise Experience and Effectiveness

Passive exercises felt smooth and comfortable. Rating (1–5): 4 Comments: _____
Active exercises felt natural and controlled. Rating (1–5): 5 Comments: _____
The range of motion felt appropriate. Rating (1–5): 4 Comments: _____
I did not feel excessive strain during exercises. Rating (1–5): 4 Comments: _____
The device helped me perform exercises effectively. Rating (1–5): 4 Comments: _____

Section E: Safety and Confidence

I felt safe while using the device. Rating (1–5): 4 Comments: _____
The device felt stable during movement. Rating (1–5): 3 Comments: _____
The emergency stop and help button was easy to reach and use. Rating (1–5): 5 Comments: _____

Section F: Device Design and Practicality

The size of the device was appropriate. Rating (1–5): 4 Comments: _____
The device did not feel too bulky. Rating (1–5): 4 Comments: _____

Section G: Effort and Workload

Using the device required too much physical effort. Rating (1–5): 3 Comments: _____
I felt comfortable completing exercises. Rating (1–5): 4 Comments: _____

Section G: Overall Feedback

What did you like most about the device? Device was intuitive to use
What did you find uncomfortable or difficult? NA
What improvements would you suggest? Having a more stable device

Would you be willing to use this device regularly if recommended by a therapist? (Yes / No) Yes

Remarks: This survey is an adapted hybrid usability survey based on recognised standards and validated constructs, refined using current lower-limb rehabilitation-device literature.

References

[1] J. Brooke, “SUS: A quick and dirty usability scale,” in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and A. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[2] L. Demers, R. Weiss-Lambrou, and B. Ska, “The Quebec User Evaluation of Satisfaction with Assistive Technology (QUEST 2.0): An overview and recent progress,” *Disability and Rehabilitation*, vol. 24, no. 1–3, pp. 21–30, 2002.

[3] S. G. Hart and L. E. Staveland, “Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research,” in *Advances in Psychology*, vol. 52, P. A. Hancock and N. Meshkati, Eds. Amsterdam, Netherlands: North-Holland, 1988, pp. 139–183.

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[6] D. R. Louie and J. J. Eng, “Powered robotic exoskeletons in post-stroke rehabilitation of gait: A scoping review,” *Journal of NeuroEngineering and Rehabilitation*, vol. 13, no. 53, 2016.

Device Comfort and Usability Survey: Patient/Client

This survey aims to evaluate the comfort, usability, and overall user experience of the stroke rehabilitation device. There are no right or wrong answers. Your honest feedback will help improve the design and usability of the device. All responses will be kept confidential.

Participant Information

Name: Nadia Rahman
Age: 22
Gender: Female
Occupation: Student
Date of Test: 3 April 2026
Tester Name: Nadia and Ron
Total Time Using Device: 15 min

Section A: General Usability

Please rate the following statements on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

I felt confident using the device. Rating (1–5): 5 Comments: _____
I would be willing to use this device regularly Rating (1–5):5 Comments: _____

Section B: Setup, Adjustment and Handling

The use process was easy to understand. Rating (1–5): 4 Comments: _____
It was easy to get into and out of the device. Rating (1–5): 4 Comments: _____

Section C: Fit and Physical Comfort

The padding and straps were comfortable. Rating (1–5): 3 Comments: _____
The device fit my leg well. Rating (1–5): 3 Comments: _____
I did not experience pressure points or pain. Rating (1–5): 4 Comments: _____
The device supported my leg securely. Rating (1–5): 5 Comments: _____
The device accommodated my posture comfortably. Rating (1–5): 5 Comments: _____

Section D: Exercise Experience and Effectiveness

Passive exercises felt smooth and comfortable. Rating (1–5): 4 Comments: _____
Active exercises felt natural and controlled. Rating (1–5): Comments: _____
The range of motion felt appropriate. Rating (1–5): 4 Comments: _____
I did not feel excessive strain during exercises. Rating (1–5): 5 Comments: _____
The device helped me perform exercises effectively. Rating (1–5): 5 Comments: _____

Section E: Safety and Confidence

I felt safe while using the device. Rating (1–5): 4 Comments: _____
The device felt stable during movement. Rating (1–5): 4 Comments: _____
The emergency stop and help button was easy to reach and use. Rating (1–5): 4 Comments: _____

Section F: Device Design and Practicality

The size of the device was appropriate. Rating (1–5): 4 Comments: _____
The device did not feel too bulky. Rating (1–5): 4 Comments: _____

Section G: Effort and Workload

Using the device required too much physical effort. Rating (1–5): 1 Comments: _____
I felt comfortable completing exercises. Rating (1–5): 5 Comments: _____

Section G: Overall Feedback

What did you like most about the device? It's quiet.
What did you find uncomfortable or difficult? The straps dug into my thigh a little bit.
What improvements would you suggest? A cushion to sit on.

Would you be willing to use this device regularly if recommended by a therapist? (Yes / No) Yes

Remarks: This survey is an adapted hybrid usability survey based on recognised standards and validated constructs, refined using current lower-limb rehabilitation-device literature.

References

[1] J. Brooke, “SUS: A quick and dirty usability scale,” in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and A. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[2] L. Demers, R. Weiss-Lambrou, and B. Ska, “The Quebec User Evaluation of Satisfaction with Assistive Technology (QUEST 2.0): An overview and recent progress,” *Disability and Rehabilitation*, vol. 24, no. 1–3, pp. 21–30, 2002.

[3] S. G. Hart and L. E. Staveland, “Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research,” in *Advances in Psychology*, vol. 52, P. A. Hancock and N. Meshkati, Eds. Amsterdam, Netherlands: North-Holland, 1988, pp. 139–183.

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[6] D. R. Louie and J. J. Eng, “Powered robotic exoskeletons in post-stroke rehabilitation of gait: A scoping review,” *Journal of NeuroEngineering and Rehabilitation*, vol. 13, no. 53, 2016.

Device Comfort and Usability Survey: Patient/Client

This survey aims to evaluate the comfort, usability, and overall user experience of the stroke rehabilitation device. There are no right or wrong answers. Your honest feedback will help improve the design and usability of the device. All responses will be kept confidential.

Participant Information

Name: Ahmad Rizqi Hidayat
Age: 21
Gender: Male
Occupation: Student
Date of Test: 3rd April 2026
Tester Name: Nadia
Total Time Using Device: 15 minutes

Section A: General Usability

Please rate the following statements on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree).

I felt confident using the device. Rating (1–5): 4 Comments: _____
I would be willing to use this device regularly Rating (1–5): 4 Comments: _____

Section B: Setup, Adjustment and Handling

The use process was easy to understand. Rating (1–5): 4 Comments: _____
It was easy to get into and out of the device. Rating (1–5): 4 Comments: _____

Section C: Fit and Physical Comfort

The padding and straps were comfortable. Rating (1–5): 4 Comments: _____
The device fit my leg well. Rating (1–5): 4 Comments: _____
I did not experience pressure points or pain. Rating (1–5): 4 Comments: _____
The device supported my leg securely. Rating (1–5): 4 Comments: _____
The device accommodated my posture comfortably. Rating (1–5): 4 Comments: _____

Section D: Exercise Experience and Effectiveness

Passive exercises felt smooth and comfortable. Rating (1–5): 4 Comments: _____
Active exercises felt natural and controlled. Rating (1–5): 4 Comments: _____
The range of motion felt appropriate. Rating (1–5): 4 Comments: _____
I did not feel excessive strain during exercises. Rating (1–5): 4 Comments: _____
The device helped me perform exercises effectively. Rating (1–5): 4 Comments: _____

Section E: Safety and Confidence

I felt safe while using the device. Rating (1–5): 4 Comments: _____
The device felt stable during movement. Rating (1–5): 4 Comments: _____
The emergency stop and help button was easy to reach and use. Rating (1–5): 4 Comments: _____

Section F: Device Design and Practicality

The size of the device was appropriate. Rating (1–5): 4 Comments: _____
The device did not feel too bulky. Rating (1–5): 4 Comments: _____

Section G: Effort and Workload

Using the device required too much physical effort. Rating (1–5): 4 Comments: _____
I felt comfortable completing exercises. Rating (1–5): 4 Comments: _____

Section G: Overall Feedback

What did you like most about the device? The device has a good user interface
What did you find uncomfortable or difficult? The device a bit shaky
What improvements would you suggest? Have more pivot points to secure the structure

Would you be willing to use this device regularly if recommended by a therapist? (Yes / No) Yes

Remarks: This survey is an adapted hybrid usability survey based on recognised standards and validated constructs, refined using current lower-limb rehabilitation-device literature.

References

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[2] L. Demers, R. Weiss-Lambrou, and B. Ska, “The Quebec User Evaluation of Satisfaction with Assistive Technology (QUEST 2.0): An overview and recent progress,” *Disability and Rehabilitation*, vol. 24, no. 1–3, pp. 21–30, 2002.

[3] S. G. Hart and L. E. Staveland, “Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research,” in *Advances in Psychology*, vol. 52, P. A. Hancock and N. Meshkati, Eds. Amsterdam, Netherlands: North-Holland, 1988, pp. 139–183.

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[6] D. R. Louie and J. J. Eng, “Powered robotic exoskeletons in post-stroke rehabilitation of gait: A scoping review,” *Journal of NeuroEngineering and Rehabilitation*, vol. 13, no. 53, 2016.